



CRYSTAL GRAPHICS DOCUMENT

Ray
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1 --- Physics

The Rendering Equation

$$L(\vec{w}_o) = L_e(\vec{w}_o) + \int_{\vec{w}_i} L(\vec{w}_i) f(\vec{w}_o, \vec{w}_i) \cos\langle\vec{w}_i, \vec{n}\rangle d\vec{w}_i \quad (1)$$

1.1 --- Model Behavior

Generally speaking, the outward radiance $L(\vec{w}_o)$ is a cumulative product of 2 **types** of inward radiance $L(\vec{w}_i)$:

- Continuous: Diffuse
- Discrete: Specular & Transmission

Although both types are well expressed in math with the same form (Equation 1), we deliberately treat them differently since this form of arithmetic expression is not easily applicable in actual computation.

Continuous Models

This type of reflection(scatter) model redirects light from a **range** of inward directions to the outward direction. As the redirection involves a continuous range, a reflection(scatter) distribution function is commonly used to approximate how much a inward direction contributes to the outward direction.

$$L_o(\vec{w}_o) = \int_{\vec{w}_i} L_i(\vec{w}_i) f(\vec{w}_o, \vec{w}_i) \cos\langle\vec{w}_i, \vec{n}\rangle d\vec{w}_i \quad (2)$$

Discrete Models

This type of reflection(scatter) model redirects light from certain inward angles. Since the inward angles are discrete (and also of a limited number in most cases), we can just add the elements together without the need to estimate integrals 🤖.

$$L_o(\vec{w}_o) = \sum_{\vec{w}_i} L_i(\vec{w}_i) f(\vec{w}_o, \vec{w}_i) \cos\langle\vec{w}_i, \vec{n}\rangle \quad (3)$$

1.2 Reflection

In the real world most materials exhibit combinations of reflection & scatter behaviors, but it is easier to discuss them separately since the underlying physical process is different and we use different approximation models.

1.2.1 Diffuse Reflection

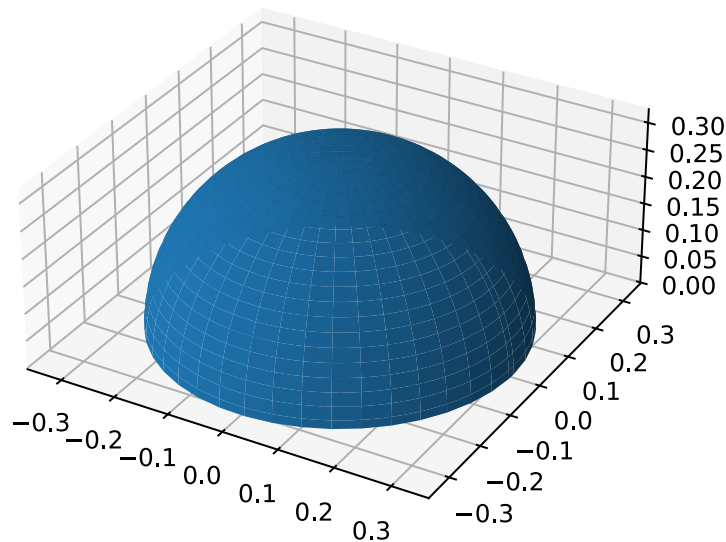


Figure 1: diffuse reflection

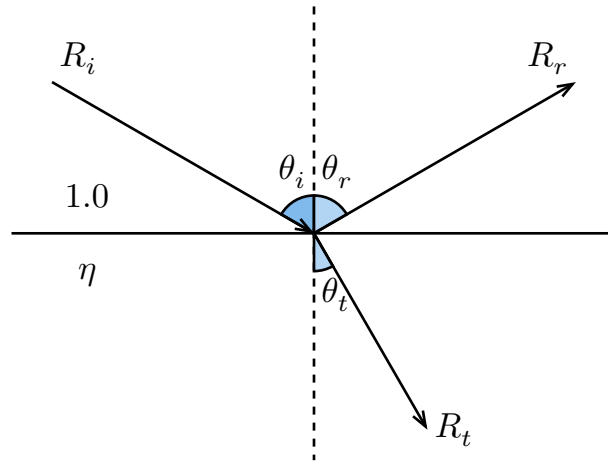
Lambertian Reflection

$$f = \frac{R}{\pi} \quad (4)$$

f is constant. And light is reflected from all inward directions in the same hemisphere.

$$\begin{aligned}
 \text{Given } R &= \iint_{\vec{w}_o} f(\vec{w}_i, \vec{w}_o) \cos \langle \vec{w}_i, \vec{w}_o \rangle d\vec{w}_o, \\
 R &= f \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \cos \varphi \sin \varphi d\theta d\varphi \quad (f \equiv C) \\
 R &= f \cdot 2\pi \cdot \frac{1}{4} \\
 f &= \frac{R}{\pi}
 \end{aligned} \tag{5}$$

1.2.2 Dielectric Specular Reflection



$$\begin{cases} \theta_r = \theta_i \\ \sin \theta_t = \eta \sin \theta_i \end{cases} \quad (\text{Snell's Law})$$

$$\begin{cases} R_r = \frac{1}{2} \left(\left(\frac{\eta \cos \theta_i - \cos \theta_t}{\eta \cos \theta_i + \cos \theta_t} \right)^2 + \left(\frac{\cos \theta_i - \eta \cos \theta_t}{\cos \theta_i + \eta \cos \theta_t} \right)^2 \right) R_i \\ R_t = R_i - R_r \end{cases} \tag{6}$$

1.2.3 Metal Reflection

1.2.4 Clearcoat

1.2.5 Sheen & Fuzz

1.2.6 Anisotropy

1.2.7 Iridescence

1.3 Transmission

1.3.1 Specular Refraction

1.3.2 Rough Refraction

1.3.3 Absorbing Transmission

1.3.4 Diffuse Transmission

1.3.5 Volume Scatter

2 Implementation

2.1 Overview

The pathtracing process is divided into the following steps:

```
0 void main:  
1   rays = camera.GenerateRays()
```

```
2   for ray in rays:
3       Render(ray)
4
5   Luminance Render(ray):
6       return Luminate(Trace(ray))
7
8   Radiance Trace(ray):
9       # 1. Hit Detection
10      hit_pos = ray.Hit(scene)
11      if not hit_pos.exist:
12          return 0
13
14      # 2. Sample Ray Generation
15      generated_rays = generate(ray, hit_pos)
16
17      # 3. Sample Tracing (Recursion)
18      for generated_ray in generated_rays:
19          radiance_in[ray] = Trace(generated_ray)
20
21      # 4. Integral Estimation
22      radiance_out = scatter(radiance_in)
23
24      return travel(radiance_out)
```

2.2 Hit Detection

2.2.1 Bounding Volumn Hierarchy (BVH)

2.2.1.1 Top Level Acceleration Structure (TLAS)

2.2.1.2 Bottom Level Acceleration Structure (BLAS)

2.3 Sample Ray Generation

The generation of sampling rays is done with the knowledge of the material of the hit position, as tracing them are **very expensive**. Knowing the material, we can make much higher quality samples with certain probability distributions

that reduce noise and make life easier for the Scatter function (the integral estimator).

2.4 GL Transmission Format (glTF)

The glTF format is used as the accepted file format. In order to perform the physics based rendering techniques from Section 1, we need to use the material information provided by the file format to determine which scatter model or combination of models to use. In addition, we also need to translate the material information to physics metrics to perform precise calculations:

glTF Material Information

- alpha mode
- anisotropy
- sheen
- base color
- clearcoat
- specular
- emission
- dispersion
- transmission
- metallic
- ior
- volume
- roughness
- iridescence

source: <https://www.khronos.org/glTF/PBR/>

We have this diagram that shows the possible combinations of active properties and the model we use for them:

Physics Behavior	Activation & Control
Diffuse Reflection	base color, roughness, metallic
(Dielectric) Specular Reflection	roughness, ior, specular
Metal Reflection	metallic, base color, specular
Clearcoat	clearcoat
Sheen	sheen
Anisotropy	anisotropy
Iridescence	iridescence
Transmission	transmission, ior, roughness
Refraction	transmission, ior, volume
Volume Absorption	volume, thickness
Dispersion	dispersion
Emission	emission

2.5 Tracing
